ACML 101   Inorganic Part

Home assignment -1

Crystal Field Theory and its applications to metal complexes

Name: ……………………………… *Entry No.* …………………………..*Group*…………

Please answer all the questions in the provided space and submit them by **December 7, 2024.**

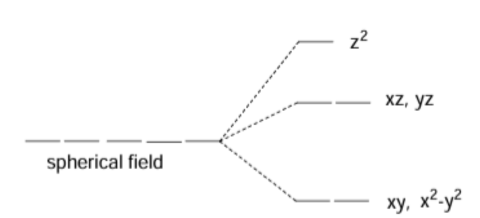
1. Calculate crystal field stabilization energy (CFSE) for the following complexes

   (a) $[Co(CN)_6]^{3-}$ (b) $[Fe(CN)_6]^{3-}$ (c) $[CoCl_6]^{4-}$

Soln.
(a) $[Co(CN)_6]^{3-}$ Co^{3+}, d^6 , CN^- strong ligand; low spin   CFSE = $-2.4\ \Delta_o + 2P$
(b) $[Fe(CN)_6]^{3-}$ Fe^{3+}, d^5 low spin   CFSE = $-2.0\ \Delta_o + 2P$
(c) $[CoCl_6]^{4-}$ Co^{2+}, d^7, Cl^- weak ligand high spin CFSE = $-0.8\ \Delta_o$

2. Draw the crystal filed splitting diagram of d orbitals in a linear complex MX_2, assuming the ligands (X) to be along the z axis.

Soln. Electron-electron repulsion between M and X will be there only for d orbitals having an z component. Out of these maximum repulsion will be for dz^2 since lobes are along the z axis. Next will be d_{xz} and d_{yz} having lobes in between z and another axis and both will be degenerate. The other two orbitals have no z component and will be stabilized to maintain barycentre.



3. Arrange $[Fe(CN)_6]^{4-}$, $[Fe(H_2O)_6]^{2+}$ and $[Mn(H_2O)_6]^{2+}$ in decreasing order of metal ion radii with justification.

Discussed in the class

4. $K_3[Co(CN)_6]$ is colorless while $K_3[Co(C_2O_4)_3]$ is colored. Why?

$K_3[Co(CN)_6]$ Co^{3+} d^6 LS $t_{2g}^6\ e_g^0$: $K_3[Co(C_2O_4)]$ Co^{3+} d^6 HS $t_{2g}^4\ e_g^2$
d-d transitions are spin allowed in both cases. However CN^- being a very strong ligand the magnitude of Δ_o will be high and absorption can occur in the UV region for the cyano complex. Therefore it is colorless.

5. Determine if $NiFe_2O_4$ is a normal or inverse Spinel?

Inverse Spinel, discussed in the class

6. Which of the following octahedral complexes are expected to show very weak tetragonal distortion and why?

(a) $[Cr(CN)_6]^{4-}$ (b) $[Mn(H_2O)_6]^{2+}$ (c) $[Cr(H_2O)_6]^{2+}$ and $[Ti(H_2O)_6]^{3+}$

Significant JT distortion means degeneracy in the e_g orbitals, discussed in the class